

Ashutosh kumar

8738878776 | ashutoshkumarm416@gmail.com | Mumbai | [Portfolio](#)

Summary:-

Full-Stack Developer experienced in **Node.js, NestJS, TypeScript, PostgreSQL, and React.js**, building scalable, secure web and mobile applications. Designed and implemented enterprise-grade **API Gateway architecture** with multi-layer security (**OAuth2/OIDC, RS256 JWT, CSRF, license enforcement**), **Keycloak-based SSO** for web and mobile clients, and microservices communication patterns. Worked on CAMS (Camera & Alert Management System) with **Kafka, WebSockets, RTSP streaming**, and RMG (Lead & Resource Management System) with RBAC and recruitment workflow management.

EDUCATION:-

GL Bajaj Institute Of Technology And Management:-

Bachelor of Technology in (Computer Science And Engineering)

Greater Noida, UP

- 2020-2024

Intermediate:-

Tomar Children School

Varanasi, UP

- 2018-2019

Matriculation:-

Sant Atulanand Convent School

Varanasi, UP

- 2017-2018

TECHNICAL SKILLS:-

- **Frontend** : React.js, JavaScript, TypeScript, **Redux Toolkit**, Tailwind CSS, Axios
- **Backend** : NestJS, Node.js, TypeScript, REST APIs, **Microservices**, API Gateway, **OAuth2/OIDC**, RS256 JWT, CSRF, **Keycloak SSO**, **RBAC**, **WebSockets**, **Apache Kafka** , Express.js, Passport.js, HTTP/HTTPS
- **Security** : RSA Licensing, Machine Fingerprinting, Replay Attack Prevention, **M2M Authentication**
- **Media & Streaming** : RTSP, HLS Streaming, FFmpeg, ROI Configuration
- **Database**: PostgreSQL, TypeORM, MongoDB, **Relational Schema Design** , SQL, Database Optimization
- **Infrastructure**: Docker, **NestJS Proxy Gateway**
- **Location Services**: GPS, Geocoding, Reverse Geocoding
- **Tools** : Git, Github , Postman, Swagger, VS Code, Cursor , Docker Compose

EXPERIENCE:-

Evolve Cortex –

Software Engineer Trainee (Full-Stack Developer)

Onsite

July 1 2025 - Present

- Contributed to **CAMS (Camera & Alert Management System)** and **RMG (Lead & Resource Management System)** using NestJS, PostgreSQL, React, TypeScript. .
- Designed **API Gateway** with multi-layer security (**License Guard → RS256 JWT → CSRF**) and **Keycloak SSO** for web and mobile authentication.
- Built User/Role/Rights/Team management APIs with Keycloak sync and RBAC.
- Implemented ROI configuration on live **RTSP** streams and QC testing module with live streaming and playback.
- Integrated **Kafka + WebSockets** for real-time SOC alert updates.
- Developed mobile **APIs** with **GPS tracking**, geocoding, image capture, and **Cron-based** auto punch-out.

PROJECTS:

1. CAMS - EYE 360 (AI-Vision)

*ReactJS | Redux T-K | Typescript | NestJS | PostgreSQL
Kafka | WebSockets | RTSP | FFmpeg | Tailwind | Docker
Keycloak | OAuth2/OIDC*

Key Features:

1. Authentication & Security Architecture -

- Designed a **single-entry API Gateway** with a multi-layer security pipeline:
License Guard → **JWT Guard (RS256/JWKS)** → **CSRF Guard** → **Proxy**, keeping **6+ microservices** isolated behind the gateway.
- Integrated **Keycloak SSO** with dual flows:
Authorization Code Flow (web) and **Direct Password Grant** (Flutter mobile app).
- Implemented **refresh token rotation** with **replay attack prevention**.
- Built **M2M authentication** using Keycloak **Client Credentials Grant** for secure service-to-service communication.
- Developed **RSA-signed software licensing** with machine fingerprinting, expiry control, clock-tamper detection, and **.env integrity** checks.

2. Backend & Microservices Development-

- Developed **Auth Service** and **Site Configuration Service** in a microservices architecture using **NestJS** with RBAC, validation, and modular design.
- Implemented Keycloak user synchronization with User CRUD, Role CRUD, Rights/Permissions, and Team management APIs.
- Built **ROI (Region of Interest)** module to draw/edit ROI on live **RTSP** camera streams for analytics and detection.
- Created **feature configuration APIs** to manage analytics features mapped to ROI areas.
- Developed **QC Testing module** with live RTSP streaming and playback to validate camera setups and alerts.
- Integrated **Apache Kafka** for sensor alerts and **Socket.IO** for real-time SOC dashboard updates.
- Implemented secure REST APIs for the **Service Engineer Mobile App** (login, logout, attendance).

3. Attendance & Field Operations Module-

- **Built Punch In / Punch Out system with:**
- GPS tracking & reverse geocoding
- Base64 image capture for proof
- Cron-based auto punch-out for inactive sessions.

2. Resource-Management-Portal

NestJS | ReactJs | NodeJs | PostgreSQL | Typescript | Redux

Key Features:

- Built a full-stack recruitment workflow system for **lead, requirement, and resource lifecycle management**.
- Implemented **JWT authentication** with access/refresh tokens and role-based access control (RBAC).
- Developed modules for **lead management, requirement creation with JD uploads, and candidate pipeline (profile → interview → placement)**.
- Integrated **file uploads (JD/resume), automated email notifications, and dashboard analytics APIs**.
- Designed relational schema with **foreign keys between leads, requirements, resources, roles, and users**.
- Built frontend with **React, TypeScript, Redux Toolkit, and Tailwind** including protected routes, form validation, and analytics dashboards.
- The system also includes **email notifications, analytics APIs, and candidate segmentation like Bench, Placed, and Released resources**.